

$$H(s) = \frac{-Z_{eq2}}{Z_{eq1}} = \frac{-R_2 // \frac{1}{sC_2}}{R_1 + \frac{1}{sC_1}} = \frac{-R_2 \cdot \frac{1}{sC_2}}{(R_1 + \frac{1}{sC_1})(R_2 + \frac{1}{sC_2})} =$$

$$= \frac{-R_2}{sC_2 \left( \frac{sC_1 R_1 + 1}{sC_1} \right) \left( \frac{sC_2 R_2 + 1}{sC_2} \right)} = \frac{-R_2}{s^2 C_1 C_2 \left( \frac{sC_1 R_1 + 1}{sC_1} \right) \left( \frac{sC_2 R_2 + 1}{sC_2} \right)}$$

$$H(s) = \frac{-sC_1 R_2}{(sC_1 R_1 + 1)(sC_2 R_2 + 1)}$$

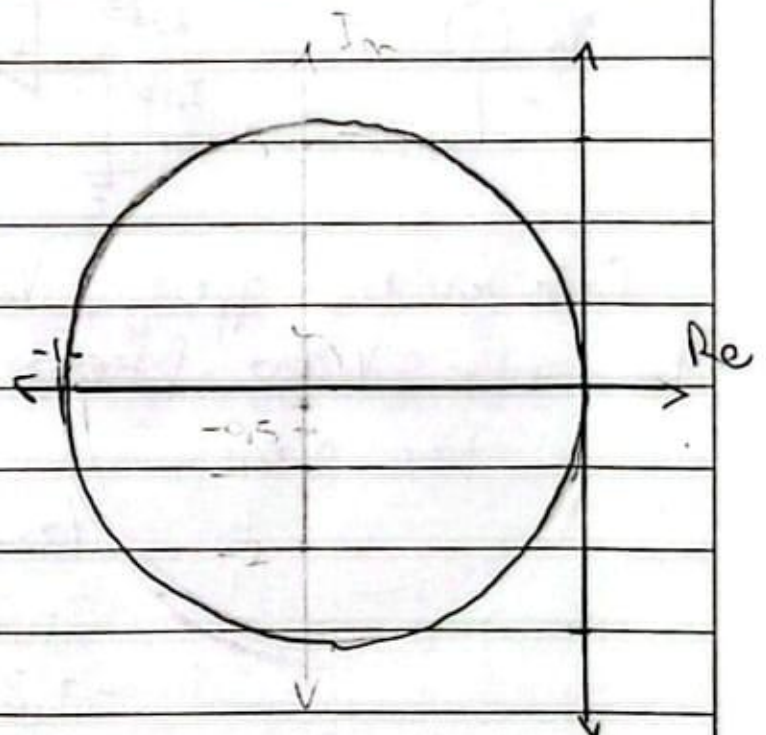
$$H(s) = \frac{-s \cdot 1,25 \times 10^{-6} \cdot 400 \times 10^3}{(s \cdot 1,25 \times 10^{-6} \cdot 20 \times 10^3 + 1)(s \cdot 15,25 \times 10^{-9} \cdot 400 \times 10^3 + 1)}$$

$$H(s) = \frac{-0,5s}{(s \cdot 0,025 + 1)(s \cdot 6,25 \times 10^{-3} + 1)}$$

Diagrama polar

$$H(j\omega) = \frac{-j\omega \cdot 0,5}{(j\omega \cdot 0,025 + 1)(j\omega \cdot 6,25 \times 10^{-3} + 1)}$$

| $\omega$ | $\text{Re}(H)$ | $\text{Im}(H)$ | $ H $ | $\angle H$ |
|----------|----------------|----------------|-------|------------|
| 0        | 0              | 0              | 0     | -90        |
| 40       | -1,46          | -4,61          | 4,84  | -107,6     |
| 40       | -11,76         | -7,06          | 13,72 | -140,4     |
| 80       | -16,76         | -10            | 16    | +80        |
| 160      | -11,76         | 7,06           | 13,72 | +140,4     |
| 320      | -4,92          | 7,38           | 8,88  | +123,7     |
| 640      | -1,46          | 4,61           | 4,84  | +107,6     |
| $\infty$ | 0              | 0              | 0     | 90         |



Valores usados en kb:

$$\hookrightarrow C_1 = (1/0,022) \mu F = 1,22 \mu F$$

$$C_2 = 15 nF$$